

Highly Efficient Perovskite Solar Cells Using Non-Toxic Industry Compatible Solvent System

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Perovskite solar cells attract a lot attention as alternative energy sources for the future energy market. With the remarkable lab-scale achievements, the investigations into a high-throughput large-scale production of perovskite devices are now on the agenda. The first step towards mass manufacturing should be a replacement of toxic solvents used in the manufacturing of perovskite layer. In this study, a non-toxic and up-scaling compatible solvent system is developed. The impact of the solvent properties on the perovskite crystallization kinetics has been systematically investigated, which is found to be crucial in controlling the surface coverage and layer crystallinity. By optimizing the processing conditions, a stabilized efficiency of 16.0% is achieved with the developed non-toxic solvent system.

1. Introduction

Nowadays, the diminishing stocks of non-renewable fossil resources and related environmental issues such as global warming and air pollution make it imperative to exploit the renewable energy technologies, taking into account the economic profitability, qualitative reliability and industrial feasibility.^[1] By converting solar energy into electrical power, photovoltaic (PV) technologies are anticipated to be one of the mainstream alternative energy resources.^[2,3] Recently, organometal trihalide perovskite solar cells (PSCs) appeared under the spotlight of PV research field, showing a steep increase in efficiency from 3.8%^[4] to a record 22% in only 6 years.^[5,6] With

the combination of advantages such as earth-abundant raw materials,^[7] ease of fabrication,^[8–10] prominent optical properties,^[11,12] and high efficiency,^[5,13] PSCs are estimated to have the shortest energy payback time among all PV technologies,^[14,15] which offers a tremendous potential for commercialization^[14] in the future PV market.

To develop a profitable perovskite technology in the context of global energy concerns, large-area coating, and printing techniques^[16–18] are desired for a high throughput production of low-cost perovskite solar cells. Before this can be achieved, however, the crystallization kinetics of perovskite must be fully understood,^[19] since the fabrication of a continuous, pin-hole free perovskite film

is crucial for high device efficiencies^[20] and yet still remains elusive. Research^[19,21,22] has revealed the strong relationship between various lead salts, solvent systems, and post-deposition treatments, which collectively govern the competition between the nucleation and growth processes during the film formation process. The realization of a fast nucleation rate at the first stage, which is followed by an extensive grain coarsening afterward is believed to be the most efficient way of producing a smooth and pinhole-free perovskite film.^[21,23,24] By using a precursor mixture of $\text{Pb}(\text{CH}_3\text{COO})_2 \cdot 3\text{H}_2\text{O}$, PbCl_2 , and $\text{CH}_3\text{NH}_3\text{I}$ via a scalable one-step spin-coating method,^[24,25] closed perovskite films can be obtained, while maintaining an excellent crystallinity. As a result, power conversion efficiencies (PCEs) of 16% have been achieved in the planar-structured perovskite solar cells.^[24]

Polar aprotic solvents such as *N,N*-dimethylformamide (DMF),^[26,27] dimethyl sulfoxide (DMSO),^[26,28] and γ -butyrolactone (GBL)^[29] are commonly used for the dissolution of lead and methylammonium salts, where they eventually regulate the film formation dynamics. Although considerable advances in efficiencies have been achieved in lab-scale perovskite devices, the commonly used DMF solvent is classified as highly toxic and volatile, which makes it unsuitable for large-scale production.^[30] The usage of a combination of methylamine and acetonitrile results in rapid, room-temperature crystallization of the $\text{CH}_3\text{NH}_3\text{PbI}_3$ precursor, making this solvent mixture a promising candidate to industrially scalable PSC manufacturing.^[31] However the very low boiling point and high flammability of this solvent system hampers its industrial uptake. Instead, GBL and

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DMSO are considered as much less harmful to the environment and potentially can be used in industry.^[29,30] However, solely GBL based inks form rough and pinhole riddled films of perovskite resulting in low PCE.^[29] High PCE can be obtained by mixing GBL with other nontoxic solvents, such as DMSO^[32] or different alcohols^[29]. Due to their nontoxic nature, these inks can be used in industry. Nevertheless, GBL is classified as a psychoactive drug, and therefore, the legal status of this solvent demands the search for other alternatives. In this light only DMSO-based inks remain the most suitable for industrial processing. DMSO is a non-toxic solvent widely used in industry and is one of the only few dipolar aprotic solvents which does not have any legislation flags required additional attention for usage in the industrial processes.^[33] Although due to easily penetration to the skin, sometimes this solvent is classified as having some safety concerns.^[29] However, the safe process for industrial use of this solvent is already established and can be utilized in the manufacturing of perovskites. It makes DMSO suitable for industrial up-scalable manufacturing of perovskite solar cells.

To achieve a pinhole-free perovskite film with high crystallinity, precursor films fabricated from pure DMSO require an elaborate and energy-consuming solvent removing procedure,^[28,34] owing to the non-volatile and coordinative solvent properties. The addition of co-solvents to DMSO is considered a promising method to fine-tune the solution properties, in particular solvents' evaporation time.^[32] While DMSO/GBL and DMSO/DMF mixtures have been widely investigated in lab-scale perovskite devices,^[28,35] no research has been done towards the utilization of a DMSO co-solvent system, which is crucial for the large-scale production of PSCs in the ambient atmosphere.

Herein, an industry compatible non-toxic solvent system is developed for the processing of perovskite films. Through the comprehensive study of Hansen solubility parameters,^[36] surface energy, volatility^[37] and the complexation of the precursor, the new combination of non-toxic solvents has shown a high solubility for perovskite precursors while maintaining an excellent wettability on the underlying substrates, a rapid drying process and, most importantly, enabling the formation of perovskite crystals.^[29] Furthermore, by changing the drying regimes at the spin-coating stage, the impact of the solvent evaporation rate on the growth of perovskite films is systematically investigated. After optimizing all processing parameters, PSCs fabricated from the new solvent system have attained a stabilized PCE of 16.0%, comparable to that of traditional DMF-based devices.

2. Results and Discussion

Searching a suitable solvent or solvent system to replace DMF or introduce co-solvents to DMSO system requires a method to predict how the perovskite precursor will dissolve. Some constraints were used to limit the set of solvents to be studied: the boiling point should not exceed the one of DMSO and the solvent should not be classified as a CMR substance (carcinogenic, mutagenic, or toxic for reproduction).

The initial approach was to apply the Hansen model^[38,39] for solubility to find solvents similar to DMF and DMSO which able to solvate the precursor. This model has been used before to find solvents for polymer solar cells^[36,40] and therefore, could be implemented to find alternative solvents for perovskites.^[29]

There are however significant structural differences between polymers and perovskite that should not be ignored. Perovskite is an organic metal hybrid material and ionic in nature, unlike polymers and most common solvents that are covalent organic chemicals. Considering that the Hansen model is designed for covalent organic chemicals, the possibility to apply this model for the ionic compounds still needs to be evaluated.

The experimental results, shown in Table S1 (Supporting Information), do not indicate any correlation between the solubility of both lead chloride (PbCl₂) and lead iodide (PbI₂) and the Hansen parameters of the solvents. Few solvents were capable of dissolving a significant amount of salt, however, these solvents do not show the shortest distance to the assumed Hansen parameters of the lead salts (the definition of assumed Hansen parameters is described in the Supporting Information). Vice versa, 2-methylpyrazine (2-MP) and GBL, which according to Hansen model should dissolve lead salts, were incapable of dissolving neither PbCl₂ nor PbI₂ on their own, leading to the conclusion that the model was not applicable.

The failure of the Hansen model most likely is attributed to the ionic nature of perovskite.^[41] There are two main attributes: solvation and complex formation,^[42] that are not incorporated in the Hansen model that strongly affect solubility of ionic compounds. Solvent molecules screen the charges via dipoles or specific interactions and create a solvation shell around the ions. Moreover, the complexation of a solvated metal ion with a ligand (complexing agent) can occur, corresponding to the substitution of the solvent molecules existing in the first solvation sphere by the molecules or ions of the ligand in solution. Thus, the metal ion–solvent interaction competes with the metal ion–ligand interaction. Solvent molecules also interact with the free ligand and the ligand coordinated to the metal ion, affecting the reactivity of the ligand. Thus, the stability of a metal complex is easily influenced by solvent.^[42]

It is well known that DMF and DMSO (Figure 1), the solvents that are the most successfully used for the deposition of perovskite, are both capable of forming complexes (adducts) in the perovskite precursor solution.^[26] To understand the solvation behavior of a solvent, possible interactions of the precursor with this solvent must be revealed. Although GBL was not considered as a suitable solvent due to legal issues, it was included in the study to better understand the solubility of perovskite precursor. Both PbI₂ and PbCl₂ are poorly soluble in this solvent. However, while neither methylammonium iodide (MAI) nor PbI₂ can be easily dissolved alone in GBL, they can be dissolved together when mixed in GBL.^[32] This indicates that MAI and PbI₂ are dissolved in GBL in the form of complex ions that interact more favorably with the solvent molecules.^[32]

The hypothesis that the Hansen model failed due to the formation of complexes between the ions in solution and their solvents was verified. For this purpose, X-ray diffraction (XRD) analysis was performed on the solid residues of the salts that did not dissolve in the solvents. These residues were dried overnight at 30 °C to ensure that no liquid solvent remained in the crystals. Drying at a low temperature was necessary to prevent a breakage the bond between ions and solvent molecules, which forms the complexes.

The analysis showed that only a few solvents caused changes in the crystal structure of lead iodide and lead chloride; all other

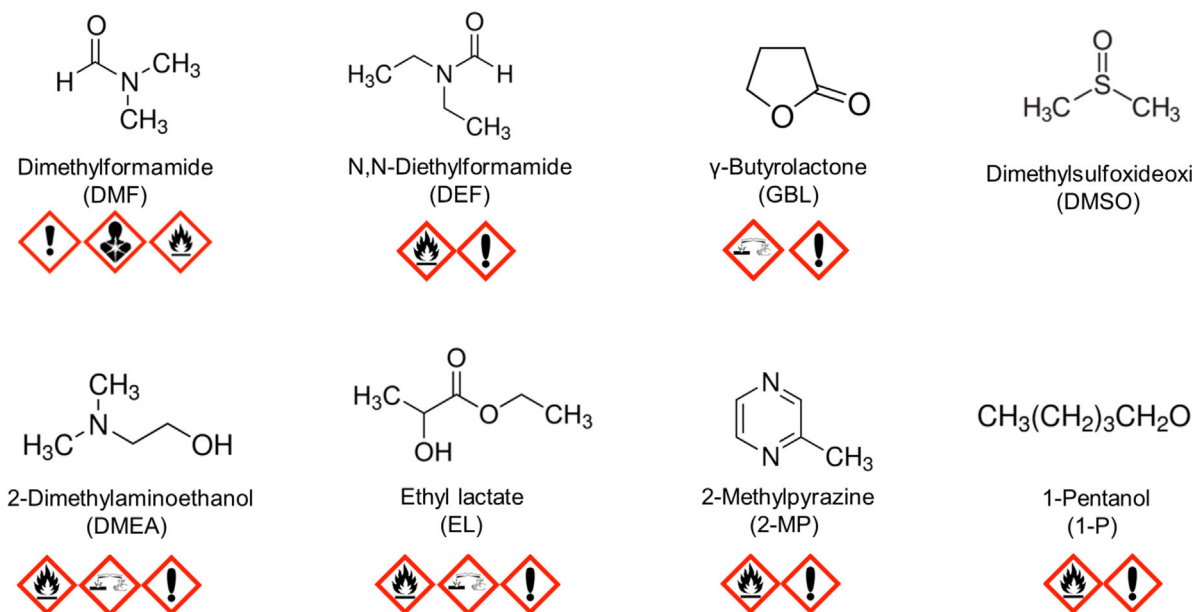


Figure 1. Chemical structures of the main solvents used in the study.

solvents show the diffraction pattern of the pure salts. The solvents that were shown to be incorporated in the crystal structure by the XRD were DMSO, 2-dimethylaminoethanol (DMAE), and 2-MP (Figure S1, Supporting Information). Moreover, lead iodide crystals formed from both DMF and *N*, *N*-diethylformamide (DEF) solutions, which did not show a complex in XRD, were different in appearance from pure lead iodide crystals before drying (Figure S1b, Supporting Information). This difference in appearance could suggest that complexes were formed, but that they were not stable at 30 °C (as already reported for DMF-based complexes^[43]). Taking this into consideration, the only solvent that appeared to be able to dissolve at least one of the lead salts, but did not seem to produce a complex, was ethyl lactate. However, considering that ethyl lactate did not reproducibly dissolve lead iodide in this study, these results cannot be taken as conclusive. This suggests that the formation of adducts with the solvent may play an active role in the solvation process of salts, which should not be ignored when looking for a solvent for a specific material.

Manufacturing of perovskite layers was successful only from solutions of DMSO, DMF, and DEF. The formation of MAPbI₃ from a 2-dimethylaminoethanol solution proved to be impossible; a different product was produced instead. Most likely this is a result of the formation of a strong complex bond between the lead ion and the solvent, which was too strong to be broken during the annealing step, or it can be considered the possibility of cation displacement with this solvent in the perovskite structure, i.e. the formation of dimethylethanolammonium lead iodide. The complexation of 2-MP with the lead salts enabled the use of 2-MP as a co-solvent added to the main solvent such as DMSO. Reducing the amount of DMSO in favor of a lower boiling solvent can be beneficial for reducing the energy needed to dry the perovskite layer. A mixture of DMSO and 2-MP in a volume ration of 1:1 was successful in the preparation of 1 M solution of both 1:1 PbI₂:MAI and 1:3 PbCl₂:MAI formulations.

Hence, two suitable candidates for replacing DMF and at least partially replace DMSO were found: DEF and the mixture of DMSO with 2-MP. DEF, which is structurally similar to DMF (Figure 1), is considered to be less harmful and is not in the CMR list. The boiling point of DEF is low enough to obtain sufficiently fast drying of the layer. However, next to solubility, the choice of solvent can significantly affect the microstructural development. For planar perovskite solar cells, the formation of a compact, smooth and closed perovskite layer will contribute to efficient light absorption and charge extraction processes that are of vital importance to the final PV performance.^[20] Upon the removal of polar solvents during the spin-coating stage, precursors gradually approach the supersaturation state where they aggregate to overcome the nucleation barrier, and small nuclei are obtained.^[21] The creation of new nuclei sites is the dominant process at ambient temperature but slows down at higher temperatures, during which the thermally activated grain growth competes with the nucleation process and thereby preferably leads to oriented crystallization.^[21] The different crystal orientation observed for DEF (Figure 2a), suggests that DEF is forming other adducts with the perovskite precursor and in this may steer crystal shape and growth, leading to a low PCE (PCE_{STAB} < 5%) (Table 1).

In the DMSO precursor solution, owing to the strongly retarded nucleation/precipitation process at room temperature, the perovskite film prepared from the instantaneous post-annealing treatment usually features by pillar-like crystals with an incomplete surface coverage (Figure 2c). In comparison, by introducing an efficient solvent quenching procedure,^[34,44,45] or by changing the solvent properties by increasing the volatility and decreasing the solubility with perovskite compounds,^[31,46] a high nucleation density can be easily achieved during the film depositing phase,^[31] which facilitates the formation of a pinhole-free perovskite layer with uniform grain sizes after the post-annealing treatment. For instance, it is found that by allowing

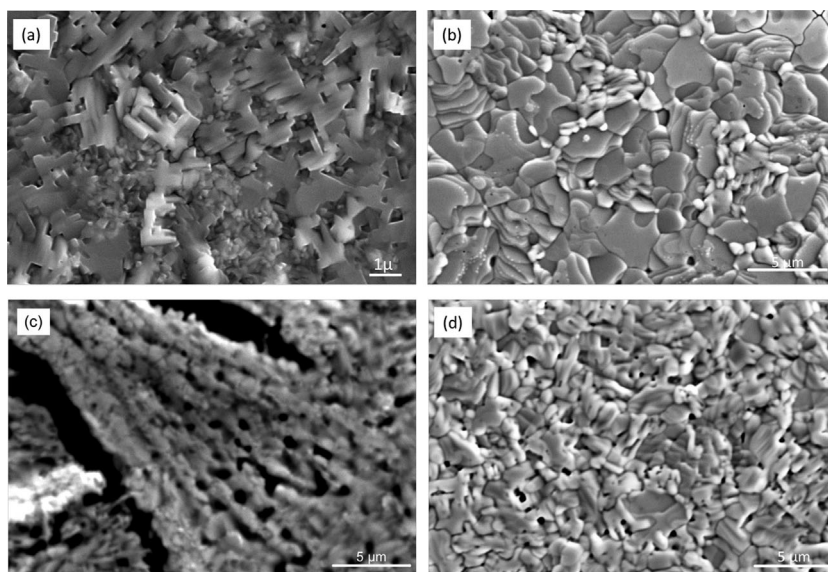


Figure 2. Surface morphology of perovskite films made from a) DEF; b) DMF; c) DMSO; and d) DMSO/2-MP/1-P (RT dried).

the precursor film to dry under a N_2 flow, the pure DMSO-based perovskite layer has shown a dense and completely covered morphology (Figure S2, Supporting Information). However, apart from the crystallization kinetics, the wettability of applied inks on the underlying substrates also affects the uniformity of perovskite films. Due to the relatively high surface tension of the pure-DMSO precursor solution (Table S2, Supporting Information), drying patterns such as surface rippling and coffee rings^[47] were often observed after the spin coating, which makes it even less applicable for large-scale coating techniques. As a result, devices made from pure DMSO solution show lower performance compared to DMF based devices (Table 1). Any other post-treatments, e.g., solvent-washing, does not help to improve the performance of the devices made from pure DMSO. The DMSO based film with RT drying is very rough and poorly crystallized, and in the absorption spectrum (Figure 3a), it shows strong light scattering below the bandgap and low absorption at short wavelengths. Both N_2 flow and ethyl acetate (EA) washed films are much smoother, but still, the absorption height is much

lower than the DMF counterpart, which is consistent with the XRD plots (Figure 3d), where a low crystallinity of pure DMSO-based film was observed. As a result, the performance of pure DMSO based devices is much lower than performance of the devices made from DMF. In addition to low performance of the devices made from pure DMSO solution, scaling up of the deposition from pure DMSO can be an issue for future industrial manufacturing. Although it was possible to spin coat perovskite precursor based on pure DMSO, large area deposition (doctor blading or slot die coating) typically results in strong dewetting and in incomplete surface coverage over the large area due to high surface tension of DMSO (Figure S3, Supporting Information). It points out that co-solvents should be added to pure DMSO solution.

It is expected that the use of lower boiling point co-solvents will help in faster solvent evaporation^[31] and reduction of surface energy, which helps in better layer formation and thereby contribute to the PV performance improvement. As shown earlier, due to the ability of 2-MP to form complexes with lead

Table 1. Photovoltaic performance of devices fabricated from different drying procedures, determined under the 100 mW cm^{-2} illumination condition. Values given in parentheses are obtained from the best performing devices.

Solvent and drying process	$V_{oc}^a)$ (V)	$J_{sc}^a)$ (mA cm^{-2})	FF ^{a)}	PCE ^{a)} (%)	PCE _{MPP} ^{b)} (300 s)
DMF	1.06 (1.09)	21.1 (21.4)	0.72 (0.75)	16.2 (17.4)	16.1
DEF	0.55 (0.57)	19.7 (20.8)	0.53 (0.56)	5.7 (6.2)	3.3
DMSO, RT dried	0.86 (0.93)	17.3 (17.2)	0.63 (0.60)	9.2 (9.7)	10.5
DMSO, N_2 dried	0.99 (1.03)	17.7 (17.6)	0.57 (0.59)	10.0 (10.8)	11.8
DMSO/2-MP/1-P, RT dried	0.94 (0.98)	21.4 (21.8)	0.62 (0.68)	12.6 (14.6)	13.8
DMSO/2-MP/1-P, N_2 dried	1.02 (1.04)	21.9 (22.0)	0.72 (0.72)	16.2 (16.5)	16.0
DMSO/2-MP/1-P, EA washed	1.00 (1.00)	21.7 (21.6)	0.69 (0.71)	14.9 (15.3)	14.5

^{a)} The average value is determined from 8–12 devices (2–3 substrates with 4 identical devices each); ^{b)} 300 s of maximum power point tracking for the best-performing devices. Here all devices from the same batch are reported, the same results were reproduced in the different batches.

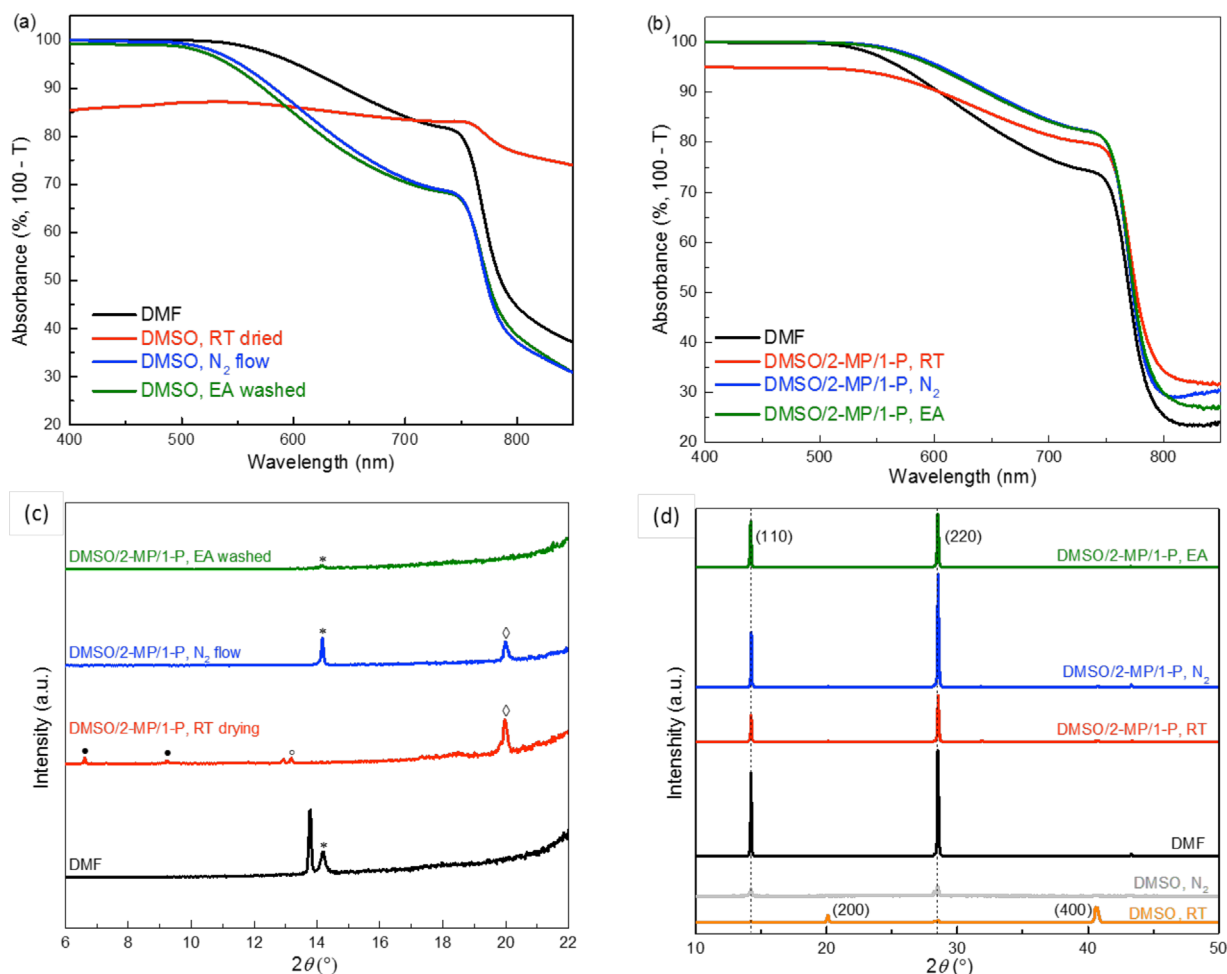


Figure 3. UV-vis absorbance of perovskite sublayers prepared from a) DMF and DMSO and b) DMF and DMSO/2-MP/1-P with different drying approaches. XRD patterns of the c) precursor and d) perovskite films on ITO/TiO₂ substrates fabricated from different solvents and drying procedures. All perovskite films were consistently annealed from 90 to 130 °C for 15 min.

salts and its good miscibility with DMSO, 2-MP is considered as a good co-solvent. Mixing DMSO and 2-MP (see the physical properties of the solvents in Table S3 of the Supporting Information) in approximately 1:1 volume ratio will not produce harmful vapors and will contribute to easier solvent removal compared to pure DMSO. Moreover, 5 vol.% of 1-pentanol (1-P) was also implemented to lower the surface energy (Table S2, Supporting Information) of the ink to allow a proper wetting of the flat TiO₂ layer. This solution was used for solar cell manufacturing, and the results were used for the comparison with devices produced from pure DMF and DMSO.

As a starting point, the new solvent DMSO/2-MP/1-P with a volume ratio of 50:45:5 was used to dissolve the Pb(CH₃COO)₂·3H₂O, PbCl₂, and CH₃NH₃I mixture at a concentration of 1 M Pb²⁺, after which the precursor was spin coated and annealed to fabricate the perovskite film. The two co-solvents are used to improve the DMSO-based ink regarding the drying regime and surface energy, respectively. Compared to the pure DMSO recipe, with the advantageous low surface energy^[36] of 1-P (Table S3, Supporting Information), the implemented solution displayed a better wettability on the underlying

substrates (glass/indium tin oxide (ITO)/TiO₂). On the other hand, similar to DMF,^[28] owing to the high volatility of 2-MP, the applied solution could rapidly become saturated after spinning and then perovskite-like crystals began to precipitate. This assumption was evidenced by the gradual transition of films from colorless to brownish upon being exposed to a static N₂ atmosphere in a glovebox for several minutes (RT drying),^[31] which significantly differs from that of pure DMSO where the transparent precursor film could be stabilized for hours. As can be seen in the X-ray diffraction (XRD) patterns of the precursor films (Figure 3c), in contrast to the DMF-based film which already exhibits a characteristic perovskite diffraction peak at 14.2°, after RT drying the DMSO/2-MP/1-P-based precursor displays diffraction peaks at small angles of 6.6° and 9.2°, which can be ascribed to the formation of the MAI-PbI₂-DMSO intermediate phase.^[35] In this case, however, no distinct perovskite peak can be identified which indicates that the formation of perovskite crystals was still within their incubation period.^[34] Therefore, by following the same post-annealing procedure, the crystallinity of the DMSO/2-MP/1-P based perovskite film is lower than that of the DMF-based film

(Figure 3d), albeit much higher than the one obtained from the pure DMSO-based perovskite films (Figure 3d). The enhancement indicates the efficient nucleation process induced by rapid salt precipitation after adding the co-solvents. Moreover, the UV-vis spectra shown in Figure 3b reveals that the film prepared from the new solvent system displays a characteristic perovskite absorbance, with an onset at around 800 nm. Also, XRD patterns shown in Figure 3d illustrates the formation of tetragonal perovskite phase, where the sharp diffraction peaks at 14.2° and 28.7° can be identified as (110) and (220) lattice planes, respectively.^[48] Both the absorbance spectra and XRD patterns have demonstrated the successful formation of a high-quality perovskite film by using the DMSO/2-MP/1-P solvent system.

It should be mentioned that, upon several minutes of RT drying, the precursor film spun from DMSO/2-MP/1-P was hazy and non-uniform, and subsequently turned into a rough perovskite layer (Figure 2d). The roughness is evidenced by the additional light absorption in the long wavelength range ($\lambda > 800$ nm, see Figure 3b) due to the scattering effect,^[49] as compared to that of the DMF-based perovskite layer. According to Figure 2d, the presence of small-sized grains manifests that the roughness is mainly caused by the uneven evaporation of solvent from the substrate surface,^[28] rather than the formation of large-sized crystals that are typical for pure DMSO inks.^[50] In comparison with the volatile DMF which rapidly dries during spin coating, the evaporation rate of the solvent mixture is not quick enough to produce a flat layer. Perovskite precursors would, therefore, preferentially precipitate at the localized liquid-vapor interface,^[22] which eventually results in an inhomogeneous distribution of small crystals.^[45]

To initiate a rapid solvent removal process and produce a uniform and a highly crystallized perovskite layer, both the inert gas flow drying,^[51] and the solvent washing^[34] processes have been investigated. In the former approach, immediately after spin coating, the wet and transparent precursor films spun from DMSO/2-MP/1-P mixture were dried under a nitrogen flow. The color change from transparent to dark brown was already observed after 30 s of N_2 blowing, and both the precursor and the perovskite film had a smooth and mirror-like appearance. In the second approach, an anti-solvent miscible with the polar solvents^[34] was dropped onto the high-speed spinning substrate, in order to wash away the excess solvent and to precipitate a uniform layer of the perovskite-DMSO complex. Here, ethyl acetate (EA), which is a non-toxic, volatile and non-polar solvent with a large dipole moment (1.78 D),^[34,52] is found to be very effective in washing DMSO. Similar to the N_2 blowing

procedure, a dark-brown color on the substrate was seen after the ethyl acetate dripping, and the subsequent annealing process produces a very uniform perovskite layer. As is evident from the UV-vis spectra (Figure 3b), compared to the RT drying, perovskite films prepared either using solvent washing approach or N_2 blowing have shown higher absorption efficiencies above the bandgap ($\lambda < 800$ nm) together with a reduced absorbance in the long wavelength range. This trend can be attributed to the fast and yet homogeneous solvent extraction which improves the layer smoothness. Perovskite films are shown to be covered with small-sized crystals (Figure 4a,b), which confirms the creation of high density of perovskite nuclei during the solvent vaporization phase. It should be noted that, compared to the compact morphology shown in Figure 2d, the porosity of perovskite films obtained from both quenching methods (Figure 4a and b) is slightly increased. However, the perovskite grains are still interconnected around the pinholes, assures sufficient light absorption in the perovskite film. Compared to the RT drying procedure, both approaches have shown to be more efficient in removing the residual DMSO, which prevents the formation of MAI-PbI₂-DMSO intermediate phase and that the perovskite crystal phase is already distinguishable before thermal annealing (Figure 3c). Furthermore, XRD patterns shown in Figure 3d demonstrate the higher crystallinity achieved in the perovskite films by introducing either of the two solvent removing procedures.

To elucidate the effect of the DMSO/2-MP/1-P solvent on the photovoltaic device performance, planar *n-i-p* PSCs based on the device structure of glass/ITO/compact-TiO₂ (c-TiO₂)/CH₃NH₃PbI_{3-x}Cl_x/2,2',7,7'-tetrakis-(*N,N*-di-4-methoxyphenylamino)-9,9'-spirobifluorene (Spiro-MeOTAD)/Au were fabricated following a similar procedure as reported by Qiu et al.^[24] The PV parameters are evaluated under the simulated one-sun illumination condition ($\approx 100 \text{ mW cm}^{-2}$) and are summarized in Table 1 (the statistical distribution of the parameters is shown in Figure S4, Supporting Information). It is remarkable that the cells processed from the new solvent system and a subsequent RT drying procedure display a average PCE of 12.6%. As is clearly seen in Table 1 and Figure S4, Supporting Information, the enhancement in PCE is mainly attributed to the increase in the short-circuit current density (J_{sc}) from 17.3 to 21.4 mA cm^{-2} , when compared to the pure DMSO-based devices. This indicates the effects of improved film crystallinity, reduced porosity, and improved connectivity between the grains, leading to more efficient light absorption in the perovskite layer. However, when compared to the DMF-based reference devices, there is a slight

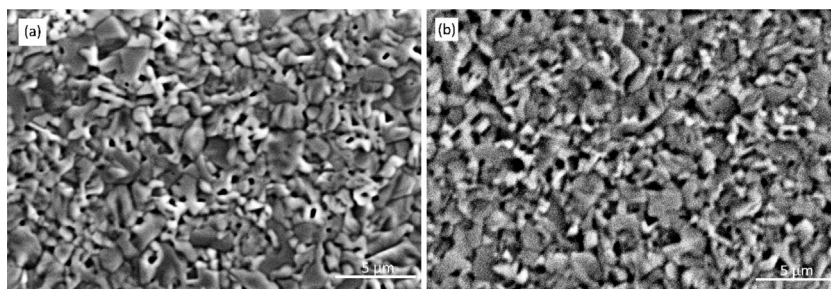


Figure 4. Surface morphology of perovskite films fabricated from either a) N_2 blow drying or b) ethyl acetate solvent washing.

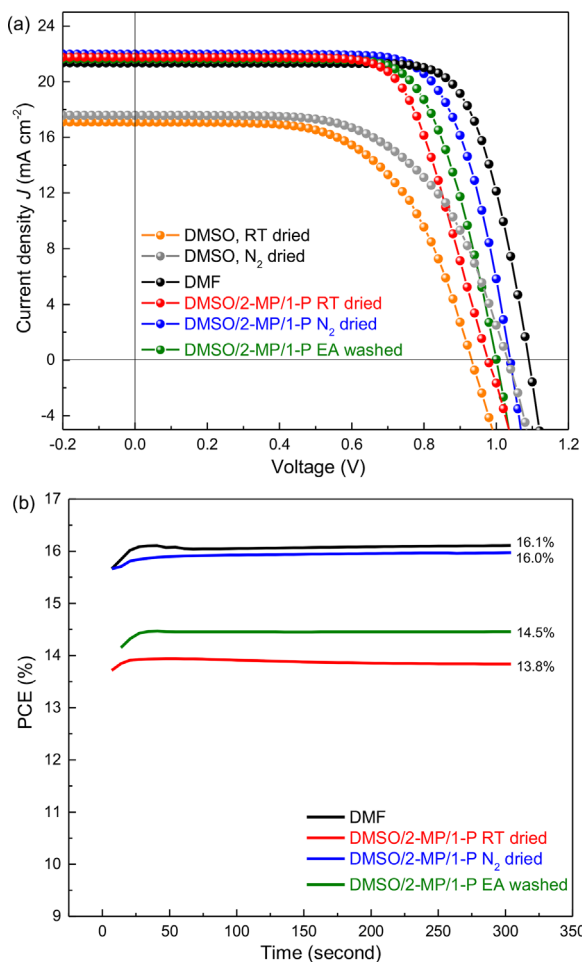


Figure 5. Under the standard light illumination, a) photocurrent plots of the best-performing cell on each substrate; b) maximum-power-point tracking of the best devices, measured under the bias V_{MPP} .

drop in the average open-circuit voltage (V_{oc}) from 1.06 to 0.94 V, and in the average fill factor (FF) from 0.72 to 0.62, which can be ascribed to either a reduced crystallinity or the higher surface roughness. As a result, more trap states are expected at the grain boundaries and substrate surfaces,^[53] contributing to the charge recombination processes that deteriorate the final PV performance. An improvement in the device performance has been achieved by changing the drying procedure after the precursor film deposition. As can be seen in Table 1, PSCs prepared from

the EA solvent washing method has attained a PCE of 14.9%, while an average PCE of 16.2% is achieved in devices by dried under a N₂ flow. In both cases, the perovskite layer smoothness as well as the crystallinity have been improved, which mitigate the excess trap states and thereby reduce the charge recombination losses. **Figure 5a** exhibits the J - V plots of the best-performing perovskite solar cells obtained from the different drying approaches. Under the 100 mW cm⁻² illumination condition, a very promising PCE of 16.5% has been achieved by using the N₂ flow drying for the DMSO/2-MP/1-P ink, which is close to the maximum PCE of the DMF-based device assembled in the same batch (17.4%). The FF and V_{oc} are still slightly lower than the DMF counterparts, due to tiny pinholes in the perovskite film. The J - V hysteresis of typical DMF-based devices and devices made from DMSO/2-MP/1-P is shown in Figure S5 (Supporting Information), and does not show any difference between these devices. Also, the power output tracking of the best devices operated at the maximum-power-point is presented in Figure 5b, which shows that the difference between DMF and N₂ dried DMSO/2-MP/1-P is negligible. Since only insignificant drops are observed in PCEs after 300 s of tracking, it is concluded that the devices prepared from the DMSO/2-MP/1-P solvent system maintain an excellent stability under the bias and light illumination condition and that the PCEs evaluated from the fast J - V scan are reliable.^[54]

Furthermore, in an attempt to further accelerate the drying process, which might be required for future roll-to-roll (R2R) processing, the possibility of usage a lower amount of DMSO in the developed solvent system has been explored. It was found that at least 35 vol.% of DMSO is needed to dissolve the Pb-(CH₃COO)₂ · 3H₂O, PbCl₂, and CH₃NH₃I mixture at a concentration of 1 M Pb²⁺. Any further decreasing the amount of DMSO turns the ink into a milky emulsion that is no longer suitable for any coating technique. On the other hand, by decreasing the concentration of Pb²⁺ to 0.37 M, a precursor solution with the DMSO content as low as 15 vol.% can still be prepared and leads to the formation of perovskite film. In decreasing the content of DMSO, perovskite solutions (DMSO/2-MP/1-P) with the volume ratios of 35:60:5 (1 M Pb²⁺) and 25:70:5 (0.5 M Pb²⁺) are further scrutinized regarding film morphology and device performance. It was observed that both recipes displayed the same drying mechanism as the 50:45:5 solution, for which the N₂ blowing treatment created the smoothest perovskite films as well as the highest PCEs. It was noted that spin coating such a diluted solution can lead to difficulties in achieving the minimum perovskite layer thickness of 300 nm, which is of vital importance to ensure the

Table 2. Photovoltaic performance of devices fabricated from different solvent compositions (N₂ blow drying), determined under the 100 mW cm⁻² illumination condition. The averaged values and standard deviation are evaluated from four devices.

	Thickness (nm)	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF	PCE (%)	PCE _{MPP} (300 s)
DMF	391	1.03 ± 0.05	20.4 ± 0.4	0.63 ± 0.01	13.3 ± 0.31	13.2
50:45:5 ^{a)}	391	0.96 ± 0.02	21.2 ± 0.3	0.71 ± 0.01	14.4 ± 0.39	14.1
35:60:5 ^{a)}	376	0.97 ± 0.03	20.9 ± 0.2	0.71 ± 0.01	14.4 ± 0.37	13.7
25:70:5 ^{a)}	242	1.00 ± 0.01	19.5 ± 0.2	0.67 ± 0.03	13.1 ± 0.77	13.0

^{a)} Volume ratio between the solvents DMSO:2-MP:1-P.

sufficient light absorption.^[11] However, by changing the rotation speed during spin coating, the perovskite layer thickness was optimized. As can be seen in Figure S6 (Supporting Information), no significant difference in the surface morphology is observed, where all the perovskite films (N₂ blowing) are uniformly covered with small-sized grains, with only a few pinholes left on top. The averaged PV parameters are summarized in Table 2 while the *J*–*V* curves are plotted in Figure S7 (Supporting Information). Under the simulated sunlight illumination condition (100 mW cm^{−2}), both devices fabricated from the 35:60:5 and 25:70:5 recipes have achieved an average PCEs of 14.4 ± 0.37% and 13.1 ± 0.77%, respectively; while the best-performing devices subsequently stabilize at 13.7 and 13.0% after 300 s of maximum-power-point tracking. The reference devices fabricated in the same batch based on DMF- and (50 vol.%) DMSO/2-MP/1-P solutions demonstrate PCE of 13.3 ± 0.31% and 14.4 ± 0.39%, respectively. It indicates that decreasing an amount of DMSO does not lead to a performance drop. Contrary to spin coating, a diluted, low concentration precursor solution (≈0.3 M) is actually preferred for coating perovskite films using large-scale coating techniques; wet and thick (5–10 μm) precursor films can be formed^[45] and dried to achieve the optimal perovskite layer thickness between 300 and 500 nm. Therefore, the developed DMSO/2-MP/1-P solvent system, which uses a very limited amount of high boiling solvents, is considered ideal for processing high-quality perovskite films, and provides a very wide process window for future R2R processing.

3. Conclusion

In summary, a solvent system compatible with industrial up-scalable manufacturing was developed for high-performing perovskite photovoltaics. Based on the industry compatible solvent mixture that provided fast solvent evaporation and good wettability, the developed non-toxic DMSO-based ink was found to initiate a rapid perovskite nucleation/crystallization process thereby contributing to the formation of compact and densely crystallized perovskite films. By introducing more efficient solvent removing approaches such as N₂ gas quenching and anti-solvent washing, a smooth and well-crystallized perovskite film was obtained. A PCE higher than 16% has been achieved after optimizing all the preparation conditions. Moreover, the continued success in fabricating highly efficient devices after reducing the content of high boiling DMSO suggests that the DMSO/2-MP/1-P solvent system is eminently suitable for future R2R production of perovskite solar cells.

4. Experimental Section

Solubility Test: In order to determine the solubility of the lead salts, 1 g of PbCl₂ or PbI₂ was put in different bottles with 3 g of solvent. The solutions were then left to stir, on a heater at 60 °C, overnight. The solutions were then removed from the hotplate and left to cool down. It was important that a solid residue was left behind to ensure that the solutions were at equilibrium and not oversaturated. The solutions were separated from the residue by passing it over a 0.5 μm PVDF syringe filter. Each filtered solution was placed in a glass container with a large open area, to facilitate evaporation. The solution was then left on a hot plate at 72 °C without stirring until all solvents evaporated. Finally, the weight of the residue was determined and compared to the weight of the solution

that was added to the bottle. Using these values, the solid contents (weight percentage) could be calculated. The results of the solubility test are summarized in Table S1 (Supporting Information).

Solution Preparation: Unless stated otherwise, all raw materials and solvents were purchased from commercial suppliers and used as received. At room temperature and an inert N₂ environment (glovebox), 1 M lead compounds (PbCl₂ and Pb(CH₃COO)₂ · 3H₂O at a molar ratio of 1:4) and 3 M CH₃NH₃I mixtures were dissolved in DMF or DMSO/2-methylpyrazine/1-pentanol solvent system and stirred at least 10 min before use. Under the same conditions, Spiro-OMeTAD was dissolved in chlorobenzene (80 mg mL^{−1}) and doped with 28.5 μL of 4-tert-butylpyridine (TBP) and 17.5 μL of bis(trifluoromethane)sulfonamide lithium salt (Li-TFSI, 520 mg mL^{−1} in acetonitrile) solutions.

Device Fabrication: A 3 × 3 cm² glass substrates with patterned ITO electrodes were purchased from Colorado Concepts. The substrates were cleaned by sequential sonication in a detergent solution, deionized water, and isopropanol. The substrates were then treated with UV-O₃ for 30 min. By using an electron beam (e-beam) PVD system (Cryofox of Polytechnik), 50 nm of c-TiO₂ was evaporated onto the ITO substrates (1 Å s^{−1}). Before depositing the perovskite layers, the substrates covered with the e-beam c-TiO₂ were further treated with 3 min of O₂ plasma, to increase the wettability of the perovskite inks. Then the 60 μL of perovskite precursor was one-step spin coated on the TiO₂ at 3000 rpm for 60 s in an N₂ filled glovebox, and the substrates were ramp-annealed from 90 to 130 °C within 10 min and further annealed for 5 min at 130 °C. After cooling down the substrates, 50 μL of the Spiro-OMeTAD solution was spin coated at 2000 rpm for 60 s onto the perovskite film, resulting in the formation of a 200 nm thick hole transporting layer. The Spiro-OMeTAD oxygen doping was attained by exposing the substrates to air overnight under controlled humidity (relative humidity = 48%). Subsequently, the substrates were transferred to the thermal evaporator under the pressure of 1.0 × 10^{−6} mbar, where a 100 nm of Au back electrode was deposited on top of the Spiro-OMeTAD film through a shadow mask. The overlap between the ITO and Au electrodes determined the active device area of 0.16 cm².

Device Characterization: In an N₂ environment, the *J*–*V* measurements were carried out under the simulated air mass (AM) 1.5G sunlight condition, using a tungsten-halogen lamp (≈100 mW cm^{−2}) and a Keithley 2400 source meter. A shadow mask 0.09 cm² was used to better define the active area. The measurements were all taken from the forward bias to negative bias (reverse scan). Top-view surface morphologies were recorded using scanning electron microscope equipment (SEM JEOL, JSM-6010LA, IntouchScope, 5–15 kV). The UV-vis absorbance was measured via Cary 5000 spectrophotometer (Agilent Technologies), by subtracting the reflection and transmission curves from the incident light. The XRD patterns of perovskite layers were carried out using an X-ray diffractometer (Panalytical X'Pert MPD), with a CuK_α radiation. Surface energy measurements were performed using EasyDrop standard Krüss instrument.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Keywords

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